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The impact of different warm-up strategies on acceleration and deceleration demands in highly trained soccer players

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Abstract

This study compared the differences in acceleration and deceleration demands between three different warm-up (WU) strategies (*Reaction speed* [exercises with reaction to a stimulus], *Run* [self-paced running], and *Speed* [exercises such as shuttle running or circuits]) applied to highly trained soccer players. Nineteen players were monitored for 4 weeks using a 10 Hz Global Positioning System. Accelerations and decelerations magnitudes were classified as low (25%–50%), moderate (50%–75%), and high (>75%) intensities. Additionally, efforts were analyzed according to their respective starting speeds (<5, 5–10, 10–15, 15–20, 20–25, and >25 km h^{-1}). Differences between WU strategies were estimated via paired mean differences along with effect sizes. The three WU strategies led to few efforts starting >15 km h^{-1} and high-intensity efforts (<1 effort per minute). Players performed more high-intensity accelerations during *Speed* than *Reaction Speed* (ES: 0.74 [90% CI: 0.21, 1.33]); more moderate-intensity accelerations during *Reaction Speed* than *Run* (ES: 1.29 [90% CI: 0.72, 2.00]); more moderate-intensity decelerations during *Reaction Speed* than *Run* (ES: 0.64 [90% CI: 0.04, 1.32]) and *Speed* (ES: 0.89 [90% CI: 0.37, 1.50]); more decelerations started at 20–25 km h^{-1} during *Speed* than *Reaction Speed* (ES: 0.81 [90% CI: 0.20, 1.49]) and *Run* (ES: 0.76 [90% CI: 0.42, 1.18]); and more decelerations started at >25 km h^{-1} during Speed than *Reaction Speed* (ES: 3.57 [90% CI: 2.88, 4.58]). In conclusion, *Speed* elicited higher acceleration and deceleration demands than the *Reaction Speed* and *Run* WU strategies, and this should be considered when designing training sessions.

KEYWORDS

athletic performance, football, speed, sprint, team-sports, training load

Highlights

- Soccer warm-ups (WUs) use different training strategies, which should be monitored independently as they elicit different acceleration and deceleration demands.

During data collection, Catarina Bajanca was Sports Scientist at Estoril Praia. During data analysis and interpretation and at the time of writing Catarina Bajanca is Sports Scientist at Sport Lisboa e Benfica.

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- Speed exercises (such as shuttle running and circuits) may expose players to more high-speed displacements and high-intensity accelerations.
- Practitioners should choose WUs strategies considering the subsequent activity to avoid under or overloading players.

1 | INTRODUCTION

The warm-up (WU) is the first activity performed in soccer before matches and training sessions. As the term suggests, WU is temperature-related and can be passive, by using external means to raise core and muscle temperature, or active, which involves specific drills that usually lead to greater benefits when compared to passive WU (Bishop, 2003a, 2003b). Despite this being common knowledge, the potential effects and demands of different WU strategies remain unclear in the current literature. Most WU effects, such as decreased stiffness, increased nerve-conduction rate, altered force-velocity relationship, and increased lactic energy provision, are temperature-related (Bishop, 2003a), and distinct WU protocols can be used to achieve a variety of physiological and neuromuscular changes. For example, a high-intensity WU can cause a greater increase in muscular temperature and induce postactivation performance enhancement for subsequent powerful soccer actions (e.g., run, jump, or shoot). On the other hand, this may also lead to faster energy depletion, which may be a detrimental factor in high-performance soccer settings (Bishop, 2003b).

Similarly, WU duration can also affect performance in soccer activities. One study (Yanci et al., 2019) compared WUs in soccer training sessions with different durations, founding improvements in sprint performance in the shorter protocol (8 min), whereas the longer protocol (25 min) induced a decrease in acceleration (ACC) capacity. The benefits of shorter protocols were also discussed in another study with the authors highlighting their time-efficiency when reporting no changes in repeated-sprint performance after shorter or longer protocols (van den Tillaar & von Heimburg, 2016).

However, the common practice among soccer practitioners appears to be distant from this evidence. For instance, using a survey of Premier League and Championship clubs, Towlson et al. (2013) reported that 89% of practitioners applied WUs with durations equal or longer than 25 min during matches. Moreover, a current Premier League club WU protocol of about 23 min had no positive effects on performance when compared with two different protocols (i.e., 3 vs. 3 small-sided games and 5 maximal repetitions in leg press) (Zois et al., 2011). Overall, WUs prior to soccer matches usually include running and mobility exercises and certain sport-specific drills, such as small-sided games, lasting around 30 min (McGowan et al., 2015). Somewhat surprisingly, stretching—static or dynamic—is also commonly prescribed in WUs, even considering the negative effects of static stretching on athletic performance (Needham et al., 2009; Silva et al., 2018).

Injury prevention is another point of concern in the prescription of WUs, an issue that has been widely discussed in the scientific literature (Al Attar et al., 2016, 2022). For this purpose, for example, the FIFA Medical and Research Center ("F-MARC") developed the FIFA 11+ WU program, a "standard" WU protocol of 20–25 min that comprises running, strength, plyometric, and balance exercises and combines cardiovascular activation and neuromuscular stimuli within the same session in an attempt to reduce and mitigate the risk of noncontact injuries in soccer players (Silva et al., 2018). Indeed, it has been observed that the implementation of FIFA 11+ as a regular practice can help to reduce the injury risk (Soligard et al., 2009) while, at the same time, can induce positive responses capable of subsequently improving soccer-specific performance (Bizzini et al., 2013).

Notwithstanding the selected WU strategy, scientific research has only focused on its effects on injuries and performance. On the other hand, training and matches have been receiving increasing attention from practitioners and researchers, specifically in relation to the load imposed on soccer players when performing these activities. This load is defined as "training load" (Soligard et al., 2016), and its monitoring can be a valuable tool to assist coaches in designing safer and more effective training programs (Robinson et al., 2017). Among a number of training load-derived metrics (e.g., running intensity, total distance, rating of perceived exertion, and heart rate variables), ACC and deceleration (DEC) emerge as some of the most frequently used training load variables for soccer training monitoring as revealed by a survey of 82 professional soccer clubs (Akenhead & Nassis, 2016). Notably, players with higher ACC capacity tend to jump higher and sprint faster over linear and multilinear trajectories (Loturco et al., 2019). Finally, the influence of ACC and DEC events on postmatch muscle damage markers, performance indicators (e.g., changes in vertical jump power output), and muscle soreness has also been investigated with interesting results in terms of correlations (e.g., evidence of moderate correlations between high accelerations and decelerations and muscle damage markers) (Hader et al., 2019). Previous research has traditionally monitored ACC and DEC demands using absolute and arbitrary thresholds (Silva et al., 2022). Nevertheless, a more recent approach suggests utilizing relative and individualized thresholds that consider the individual starting speed of each player (Silva, Serpiello, et al., 2023c). By employing this latter strategy, it becomes possible to report ACC and DEC demands based on the individual characteristics of each player within a real-world scenario.

Considering the aforementioned rationale, monitoring ACC and DEC demands across different WU strategies, commonly prescribed before soccer-specific training sessions and matches, may be extremely relevant from a practical perspective. To the best of our knowledge, this has not been previously investigated. Indeed, the potential differences between playing positions during different WU strategies have yet to be thoroughly discussed. Therefore, the purpose of this study was to compare the ACC and DEC demands of soccer players according to their playing positions during different training WU strategies.

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2 | METHODS

2.1 | Participants

Data were obtained from one team playing in the Portuguese U23 League (Revelation League) across 19 training sessions (full mesocycle). The participants were 33 team members, classified as "highly trained subjects" (i.e., team-sport athletes competing in national or state tournaments) (McKay et al., 2022). The sample size was determined using the G-Power software and the calculation resulted in a total sample size of 19 participants, providing an actual power of 0.81. The exclusion criteria for this study included participants who had <50% participation in training sessions, regardless of the reason, and goalkeepers, as their training routine was specific to their position. A total of 19 players (20.1 ± 1.2 years; 179.9 ± 4.5 cm) were included in the study. Players were divided according to their playing' positions as follows: 3 central defenders, 4 fullbacks (FB), 6 central midfielders (CM), 4 wide midfielders (WM), and 2 forwards. The study was conducted in accordance with the Declaration of Helsinki, and Ethics Committee approval was obtained (35/2021).

2.2 | Procedures

In this cross-sectional study, players used Global Positioning System (GPS) equipment as required for training routine monitoring and accepted data sharing for research purposes. Collected data was not filtered by the equipment software, and each player was analyzed individually. From velocity and time data, we computed the ACC and DEC demands and the time spent in these tasks for the relative count (i.e., number of efforts per minute). We also collected the starting speed of each effort, for both ACC and DEC events.

WUs were divided into three different strategies, as previously defined by the coaching staff, without any interference or adaptation for research purposes. Across 19 training sessions, all players performed the same WU, but the WU strategy varied (for all players) according to the training sessions. WU strategies were: (a) *reaction speed WU* (8–15 min; performed during sessions the day before the match): after activation drills, players performed drills with players competing with each other, where players needed to react as quick as possible after a given stimulus (agility development, including perceptual and decision-making ability); (b) *run WU* (9–14 min; performed during sessions the day after the match): after activation drills, players ran at low-intensity speeds, in a self-selected pace, but with instructions to avoid high speeds; and (c) *speed WU* (14–20 min; performed during sessions 2 days before the match): after activation drills, players performed high-intensity drills, such as shuttle runs or circuits, organized in competitive and noncompetitive formats, focusing on acceleration, deceleration, and speed efforts preparing players for the following drills (where players were able to achieve >90% maximal speed). Players were required to perform different WUs without pausing to rest or drink fluids. Specific motivation or feedback was only provided when players were not meeting the strategies objectives.

GPS units (Catapult S7; Catapult Sports) sampling at 10 Hz were used to track players during the WU sessions. The GPS unit was placed on the upper middle back, between the scapulae of the player, using the special protective vests recommended by the manufacturer. This equipment is a certified FIFA Electronic Performance and Tracking System. Each effort was counted from the start of the speed increase or decrease until the change in speed reached 0 m s^{-2} (Silva, Nakamura, Ribeiro, et al., 2023b). ACC and DEC intensities were established relative to the maximal effort registered during the mesocycle by each player based on the "percentage intensity approach" (Silva, Serpiello, et al., 2023c) adapted from the method proposed by Sonderegger et al. (2016) at 25%–50% (low intensity), 50%–75% (moderate intensity), and >75% (high intensity). To determine the maximal effort, we collected the highest ACC and DEC values of the mesocycle of each player, excluding isolated values (two efforts must have occurred in the same m s^{-2} bandwidth interval). For example, if the maximal ACC value was 7.30 m s^{-2}, another value between 7 and 8 m s^{-2} had to exist for this value to be considered as "maximal." Additionally, the starting speed of each ACC and DEC event was also collected according to bandwidth intervals: (<5, 5–10, 10–15, 15–20, 20–25, and >25 km h^{-1}) (Silva, Serpiello, et al., 2023c). The number of efforts per minute that occurred during each WU strategy was counted according to their intensity and starting speed.

2.3 | Statistical analysis

Means ± standard deviations were calculated for all analyzed variables, including the multiple ACC and DEC intensities, and starting speed intervals. Paired mean differences were computed with the Jamovi software (Version 2.3.19.0; JAMOVI project, 2022) using the ESCI package (Jamovi, n.d.; Team, 2021). Two different measures were analyzed separately: the number of efforts of one player during each WU condition was compared to the number of efforts of the same player in the remaining conditions. This procedure was conducted for each effort during the respective WUs and for the different ACC and DEC intensities and starting speed intervals. Effect sizes (ES) were established as trivial (<0.2), small (0.2 < 0.6), moderate (0.6 < 1.2), large (1.2 < 2.0), very large (2.0 < 4.0), and huge (>4.0) with 90% confidence intervals (Hopkins et al., 2009). If the CI crossed zero, an unclear effect size was established (Batterham & Hopkins, 2006). Playing positions were compared on number of ACC and DEC per minute, according to the WU protocol. Differences were analyzed with independent group contrasts, with 90% CI, also using the ESCI package (Jamovi, n.d.; Team, 2021). If the 90% CI crossed zero, differences were considered unclear.

3 | RESULTS

Table 1 presents the number of efforts per minute as mean [90% CI] according to the WU strategy. Figure 1 presents the box plot of number of efforts per minute according to their respective

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TABLE 1 Mean [90% confidence interval] of efforts per minute according to the intensity and starting speed for each warm-up strategy.

	Reaction	Run	Speed
ACC 25%–50%	1.36 [1.21, 1.50]	1.14 [0.96, 1.33]	1.05 [0.95, 1.16]
ACC 50%–75%	0.19 [0.16, 0.22]	0.08 [0.05, 0.11]	0.14 [0.10, 0.18]
ACC >75%	0.01 [0.00, 0.02]	-	0.04 [0.02, 0.05]
DEC 25%–50%	0.72 [0.56, 0.89]	0.32 [0.22, 0.43]	0.54 [0.39, 0.69]
DEC 50%–75%	0.02 [0.01, 0.03]	0.00 [0.00, 0.01]	0.00 [0.00, 0.01]
DEC >75%	-	0.00 [0.00, 0.01]	0.00 [0.00, 0.00]
ACC <5 km h^{-1}	30.59 [29.70, 31.49]	29.68 [27.00, 32.37]	33.84 [32.57, 35.11]
ACC 5–10 km h^{-1}	12.00 [10.77, 13.22]	10.15 [8.17, 12.12]	9.50 [8.07, 10.94]
ACC 10–15 km h^{-1}	0.17 [0.12, 0.23]	0.30 [0.13, 0.47]	0.11 [0.06, 0.16]
ACC 15–20 km h^{-1}	0.02 [0.01, 0.03]	0.03 [0.01, 0.05]	0.01 [0.00, 0.02]
ACC 20–25 km h^{-1}	-	-	0.02 [0.00, 0.04]
ACC >25 km h^{-1}	-	-	0.08 [0.06, 0.10]
DEC <5 km h^{-1}	26.10 [25.09, 27.12]	25.74 [22.99, 28.48]	30.58 [29.13, 32.03]
DEC 5–10 km h^{-1}	16.69 [15.26, 18.11]	14.95 [13.08, 16.82]	13.85 [12.44, 15.25]
DEC 10–15 km h^{-1}	0.71 [0.60, 0.83]	0.77 [0.52, 1.02]	0.35 [0.29, 0.42]
DEC 15–20 km h^{-1}	0.28 [0.24, 0.32]	0.18 [0.13, 0.23]	0.26 [0.22, 0.29]
DEC 20–25 km h^{-1}	0.03 [0.01, 0.04]	0.01 [0.00, 0.02]	0.08 [0.05, 0.11]
DEC >25 km h^{-1}	0.00 [0.00, 0.00]	-	0.24 [0.21, 0.28]

Abbreviations: ACC, acceleration; DEC, deceleration.

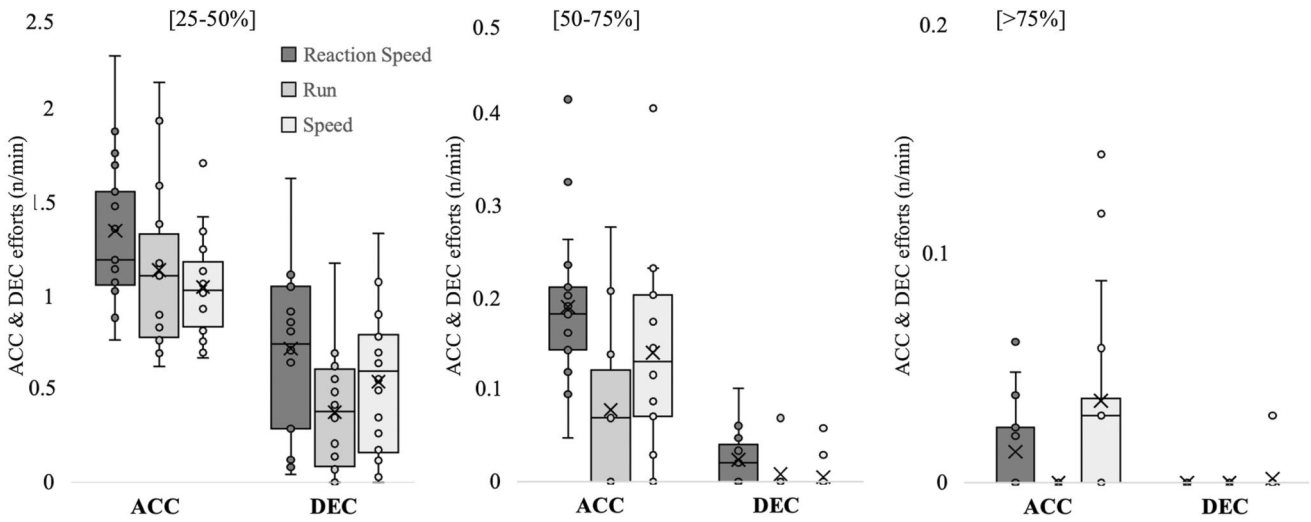

FIGURE 1 Box plot of accelerations and decelerations per minute for each WU strategy during distinct intensity intervals. ACC, acceleration; DEC, deceleration; WU, warm-up.

intensities. The *Reaction speed WU* elicited more efforts than the others WU strategies at low and moderate intensities. Less than 1 effort per minute were registered at moderate and high intensities, and more ACC were registered than DEC except high-intensity DEC during the *Run WU*.

Figure 2 presents the box plot of the number of efforts per minute according to starting speed intervals. Efforts that started at higher speeds (>20 km h^{-1}) occurred almost exclusively during the *Speed WU*.

Figure 3 presents the effect sizes found in paired mean differences between WU strategies. CI can be asymmetrical as the upper margin of error can be smaller or larger than the lower margin error (Cumming & Calin-Jageman, 2016). Effect sizes are displayed for efforts intensities and efforts starting speed.

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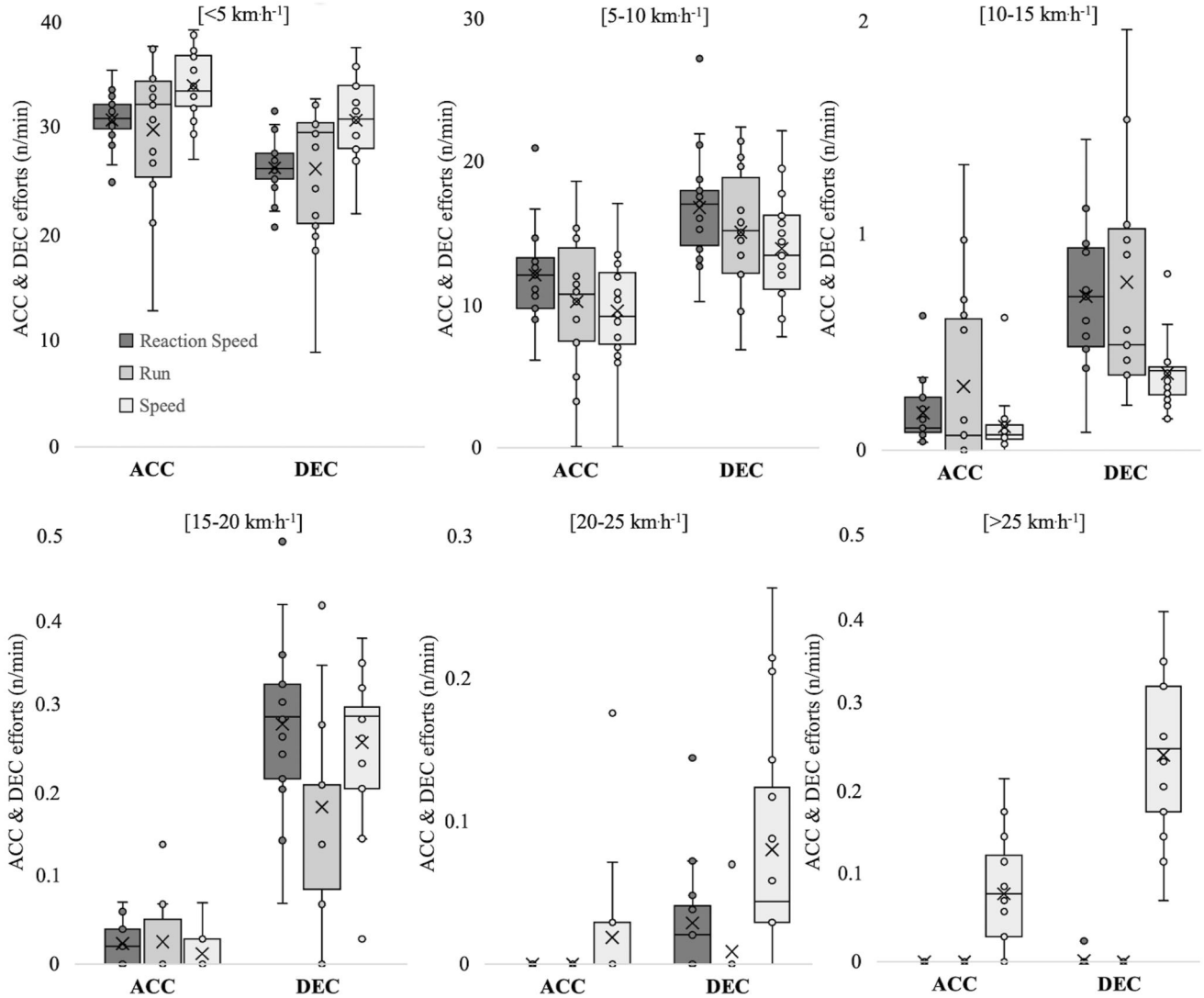

FIGURE 2 Box plot of accelerations and decelerations per minute according to the starting speed (<5, 5–10, 10–15, 15–20, 20–25, and >25 km h^{-1}) of each effort during different WU strategies. ACC, acceleration; DEC, deceleration; WU, warm-up.

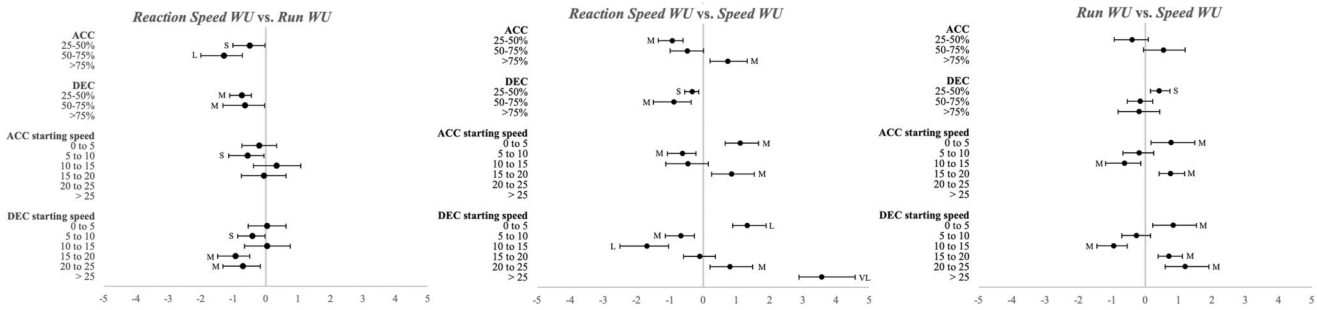

FIGURE 3 Effect sizes (with 90% confidence interval) of the mean differences between accelerations and decelerations per minute according to the respective intensity percentage and starting speed (km h^{-1}) intervals. ACC, acceleration; DEC, deceleration; L, large effect size (1.2 < 2.0); M, moderate effect size (0.6 < 1.2); S, small effect size (0.2 < 0.6); T, trivial effect size (<0.2); VL, very large effect size (2.0 < 4.0); WU, warm-up. No comparison is available if no effort is registered in a specific interval.

3.1 | Effort intensity

Players performed more low and moderate ACC (ES: 0.49 [0.02, 1.01]; ES: 1.29 [0.72, 1.29], respectively) and DEC efforts (ES: 0.74 [0.45, 1.11]; ES: 0.64 [0.04, 1.32], respectively) during the *Reaction WU* than during the *Run WU*. Players performed more low ACC and DEC efforts (ES: 0.94 [0.60, 1.36]; ES: 0.33 [0.13, 0.56], respectively) and more moderate DEC efforts (ES: 0.89 [0.37,

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1.50]) during the *Reaction WU* than during the *Speed WU*. Players performed more high ACC efforts (ES: 0.74 [0.21, 1.33]) during the *Speed WU* than during the *Reaction WU* and more low DEC efforts (ES: 0.42 [0.15, 0.74]) during the *Speed WU* than during the *Run WU*.

3.2 | Effort starting speed

More ACC events were registered during the *Reaction WU* than during the *Run WU* at (starting speed of) 5–10 km h^{-1} (ES: 0.56 [0.05, 1.14]). More DEC events were registered during the *Reaction WU* than during the *Run WU* at 5–10 km h^{-1} (ES: 0.42 [0.03, 0.87]), 15–20 km h^{-1} (ES: 0.93 [0.49, 1.48]), and 20–25 km h^{-1} (ES: 0.70 [0.17, 1.32]). More ACC events were registered during the *Reaction WU* than during the *Speed WU* at 5–10 km h^{-1} (ES: 0.62 [0.21, 1.08]). More DEC events were registered during the *Reaction WU* than during the *Speed WU* at 5–10 km h^{-1} (ES: 0.67 [0.27, 1.14]) and 10–15 km h^{-1} (ES: 1.70 [1.05, 2.51]). More ACC events were registered during the *Speed WU* than during the *Run WU* at 0–5 km h^{-1} (ES: 0.78 [0.17, 1.48]) and 15–20 km h^{-1} (ES: 0.76 [0.42, 1.18]). More DEC events were registered during the *Speed WU* than during the *Run WU* at 0–5 km h^{-1} (ES: 0.83 [0.22, 1.54]), 15–20 km h^{-1} (ES: 0.71 [0.39, 1.11]), and 20–25 km h^{-1} (ES: 1.19 [0.59, 1.91]). More ACC and DEC events were registered during the *Run WU* than during the *Speed WU* at 10–15 km h^{-1} (ES: 0.63 [0.13, 1.20]; ES: 0.95 [0.54, 1.46], respectively).

Differences between playing positions were only found in the *Run WU* with CM (11.2 [2.31, 20.09]) performing more ACC and WM (10.81 [1.52, 20.10], 10.42 [0.63, 20.20]) performing more ACC and DEC events per minute than FB.

4 | DISCUSSION

The purpose of this study was to assess and compare the ACC and DEC demands imposed by three distinct WU strategies in highly trained soccer players according to their playing positions. Overall, across all conditions, we observed that efforts with lower intensities occurred more frequently than efforts with higher intensities, similar to what has been systematically reported for soccer-specific training sessions (Silva et al., 2022). Specifically, the *Speed WU* (the WU activity with the longest duration [14–20 min] among the four WU strategies) elicited, on average, at least one high-intensity effort (ACC) per session and led to greater number of efforts starting at higher speeds. The lowest demands were clearly observed in the *Run WU*, where no high-intensity ACC or any effort above 25 km h^{-1} occurred.

Our findings present a novel alternative to monitoring that assesses training sessions as a whole. In fact, when monitoring the training load, sport scientists usually consider the demands occurred during the entire training session (categorized according to the distance in days to the match day) and not the demands of each training session drill or section (Silva, Nakamura, Castellano, et al., 2023a). However, this strategy potentially underestimate WU loads due to the reduced time spent in this activity in comparison to the rest of the training session. This is important because load differences between specific-soccer exercises exist (Silva et al., 2022), including between WU and other soccer-activities (e.g., matches, small-sided games, technical-tactical exercises, etc.), as showed by Ramos and colleagues (Passos Ramos et al., 2019) in female soccer players. Therefore, sport scientists are encouraged to monitor training activities independently and also considering the relation between activities in addition to the relation between consecutive training sessions. Finally, the lack of significant differences between distinct playing positions may be explained by the fact that all players performed the same activities; hence, the differences found during the *Run WU* are probably associated with intersubject variability in self-selected pace.

If practitioners select the *Speed WU*, higher demands may be expected, especially for efforts starting at speeds >20 km h^{-1}. Since this exercise is characterized by intense movements, such as shuttle runs, players will potentially achieve higher running speeds, which consequently leads to more intense ACC and DEC events. Indeed, shuttle running was previously compared to 1 versus 1 small-sided game with same duration, eliciting fewer ACC and DEC events >3 m s^{-2}, but with higher distance covered at speeds >25.2 km h^{-1} (compared to the game format) (Ade et al., 2014). However, when compared to a small sided game that included a higher number of players (6 vs. 6), shuttle running led to more time spent in ACC and DEC efforts (>1.5, >2.5, and >3.5 m s^{-2}) (Stevens et al., 2015). Therefore, during training sessions, players can potentially be exposed to high ACC and DEC loads if WU drills such as *Speed* are disregarded. In summary, among all tested WUs, the *Speed WU* was the strategy where players reached higher speeds and where more high-intensity efforts occurred; therefore, this strategy should be avoided during training sessions close to competition in order not to compromise match preparedness and/or recovery between successive matches (Silva, Nakamura, Castellano, et al., 2023a). Due to its high intensity, the *Speed WU* may increase substantially energy depletion (Stevens et al., 2015); nonetheless, with adequate recovery and appropriate prescription (e.g., short duration), this WU strategy can also be used to acutely improve physical performance in training and competition (Bishop, 2003b). With this in mind, incorporating the *Speed WU* during training sessions with moderate ACC and DEC loads would likely be more suitable.

The *Reaction Speed WU* elicited more low and moderate intensity ACC and DEC events per minute than the other two WU strategies. Regarding starting speed intervals, the *Reaction Speed WU* led to more ACC efforts starting between 5 and 10 km h^{-1} and between 15 and 20 km h^{-1}. Generally speaking, during this type of WU, players perform more high-intensity ACC actions than in the other two WUs. However, the data obtained during the *Speed WU* highlight the role played by movement repetition in training load. Although the *Reaction Speed WU* increased the relative frequency of high-intensity ACC efforts, this WU strategy prioritizes players' reaction. In this regard,

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the rest between successive efforts during the *Reaction Speed WU* is longer than, for example, during the *Speed WU*. In fact, in the *Reaction Speed WU*, after performing a very intense effort in response to a given stimulus, players returned to their previous position slowly in order to allow adequate recovery. In contrast, during shuttle drills, players are usually requested to quickly return to their previous (initial) position. Since directional changes increased the time spent in ACC maneuvers (Akenhead et al., 2015), this quick return to "the base-line" could be one of the major reasons to justify the difference in physical demands. Based on the previous considerations, the *Reaction Speed WU* could be included in various training sessions as long as players are not required to rapidly return to the starting point. This simple adjustment would effectively prevent excessive DEC demands on the players, thus reducing the risk of overloading during the training week and increasing the injury risk (Harper & Kiely, 2018). As expected, the *Run WU* was the least demanding WU among all analyzed strategies. Interestingly, during the *Run WU*, we observed more efforts occurring between 10 and 15 km h^{-1} than during the *Speed WU*, which means that players performed this activity predominantly within this respective speed range. Straight running at constant and comfortable speed will certainly result in lower muscular and metabolic costs (e.g., oxygen consumption) as compared to more intense and sport-specific activities, such as linear and/or multidirectional sprints (Dolci et al., 2018). For this reason, this type of activity is commonly used not only as a WU routine but also as a recovery strategy for soccer players (Rey et al., 2018).

Since different workout durations affect players in a different way (Yanci et al., 2019), distinct WU strategies should also receive attention from practitioners and researchers. Even if the main goal of WU is to prepare athletes for the following activities (i.e., training session or match), previous research recommended progressively increases in intensity until maximal efforts occur close to the end of the WU (Silva et al., 2018). Under this perspective, WU can also help athletes improve (or at least maintain) their physical performance during the competitive season. At first glance, implementing shuttle run or reaction speed drills may concern coaches, who might believe that those drills could increase the risk of injury and level of fatigue. However, a solid body of evidence confirms the effectiveness of intense WU routines in acutely improving physical performance (Zois et al., 2015) and reducing the risk of injury (van Beijsterveldt et al., 2013). Interestingly, soccer players that covered higher distances in high-speed running (>14.4 km h^{-1}) and sprinting (>19.8 km h^{-1}) were at reduced noncontact injury risk than their counterparts (Malone et al., 2018) probably because these players are better prepared to cope with match demands (Ehrmann et al., 2016). Moreover, exposing players to chronic higher loads may be crucial to ensure their fitness preparedness, minimizing the noncontact injury risk, as long as that exposure occurs without sudden and excessive load spikes (Bowen et al., 2020). Thus, WU is a part of training that can have positive chronic effects on football players, and this will probably depend on the characteristics of each WU session.

Our findings present a step toward understanding the effects of certain WU strategies on training load and potential training adaptations. First, we compared the ACC and DEC demands of different, but commonly used, WU strategies; second, we compared these demands between different playing positions. This study presents two major limitations: (1) these results refer to a particular soccer team with a particular coaching staff responsible for prescribing and monitoring WU activities; therefore, the extrapolation of our findings to other populations (including soccer clubs) should be made with caution; (2) the investigated WUs are inserted into distinct strategies; thus, they do not have a fixed or programmed structure, such as the "FIFA 11+," which compromises the replication of this research and the prescription of these strategies.

In conclusion, all WU strategies led to very few high-intensity efforts, especially high-intensity DEC. Additionally, the different WU strategies elicited different demands, highlighting the importance of considering and examining WU activities in a separate way. During the *Speed WU*, players achieved higher speeds, which resulted in a higher number and frequency of high intensity efforts and, therefore, of intense ACC and DEC. These differences were even more pronounced when compared to the *Run WU*, which was found to be the least demanding WU routine among the three analyzed strategies. The *Reaction Speed WU* led to more low and moderate intensity ACC and DEC events per minute than the other two WU strategies, a fact that can be easily explained by considering the longer intervals (and adequate recovery) typically provided during "reactive drills." Future studies are needed to test and compare the effects of using different WU strategies under similar training and competitive conditions. Coaching staff should be aware that WU can play a critical role in promoting training responses and adaptations. When selecting different WU strategies, practitioners should always consider the main objective of the training session. For example, the *Reaction Speed WU* may be applied before technical-tactical training sessions where rapid responses and reactions are expected. In contrast, the *Speed WU* may be used prior to high-intensity sessions, where a higher number of intense ACC and DEC efforts usually occur, in order to prepare players more specifically for these complex demands. The *Run WU* seems to be more indicated for preparing players for longer and/or moderate training sessions, or even as a re-WU strategy during official soccer matches. Regardless of the WU strategy, it is essential to monitor the starting speed of repeated efforts, as it has a crucial impact on ACC and DEC demands. Finally, coaches should consider that WU routines, whether implemented acutely or chronically, could potentially affect training load and should therefore be assessed and analyzed separately.

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CONFLICT OF INTEREST STATEMENT

There is no conflict of interest to declare.

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